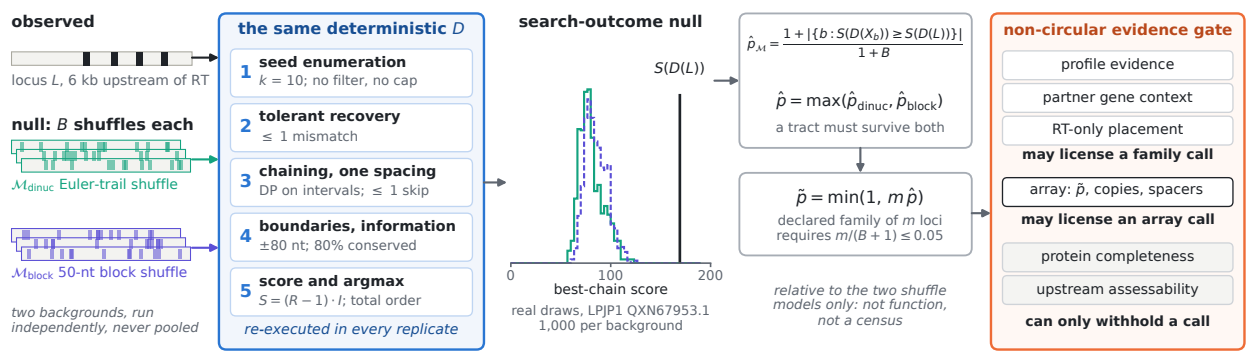
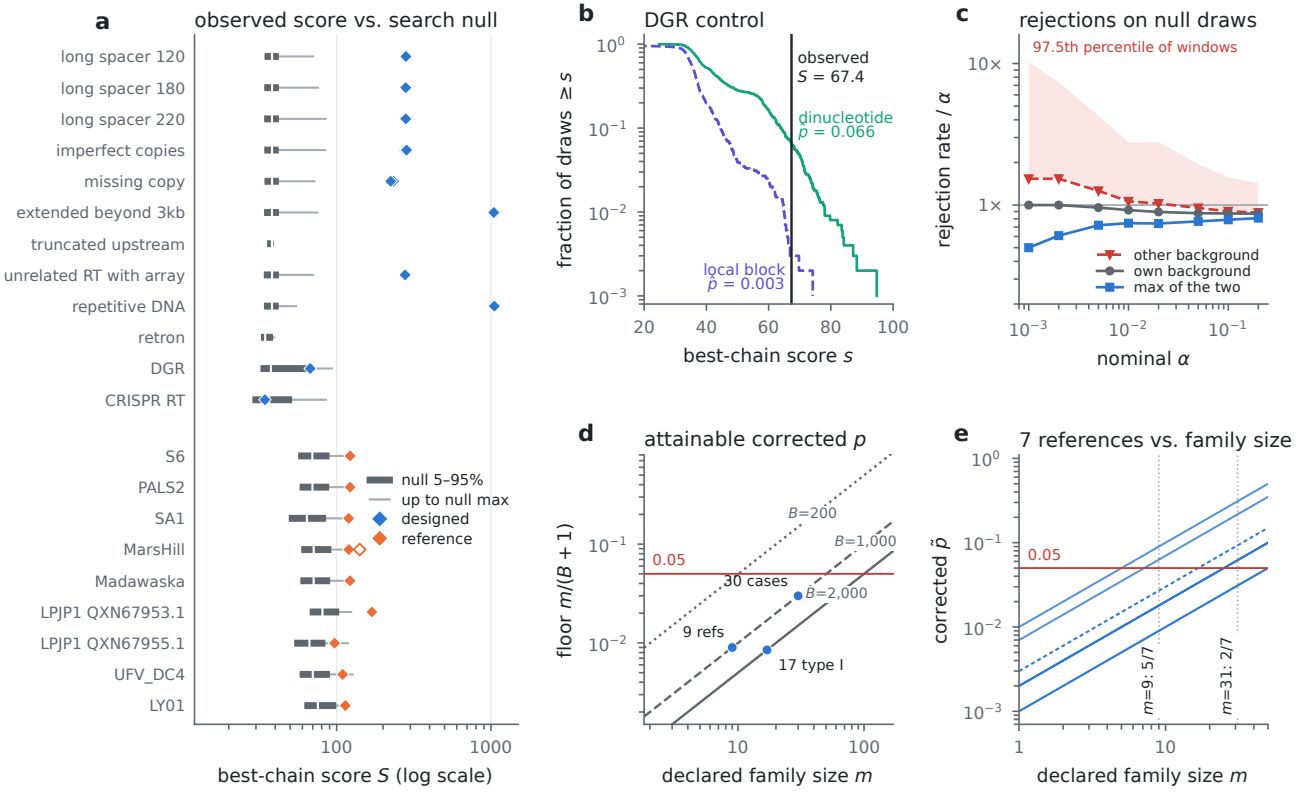
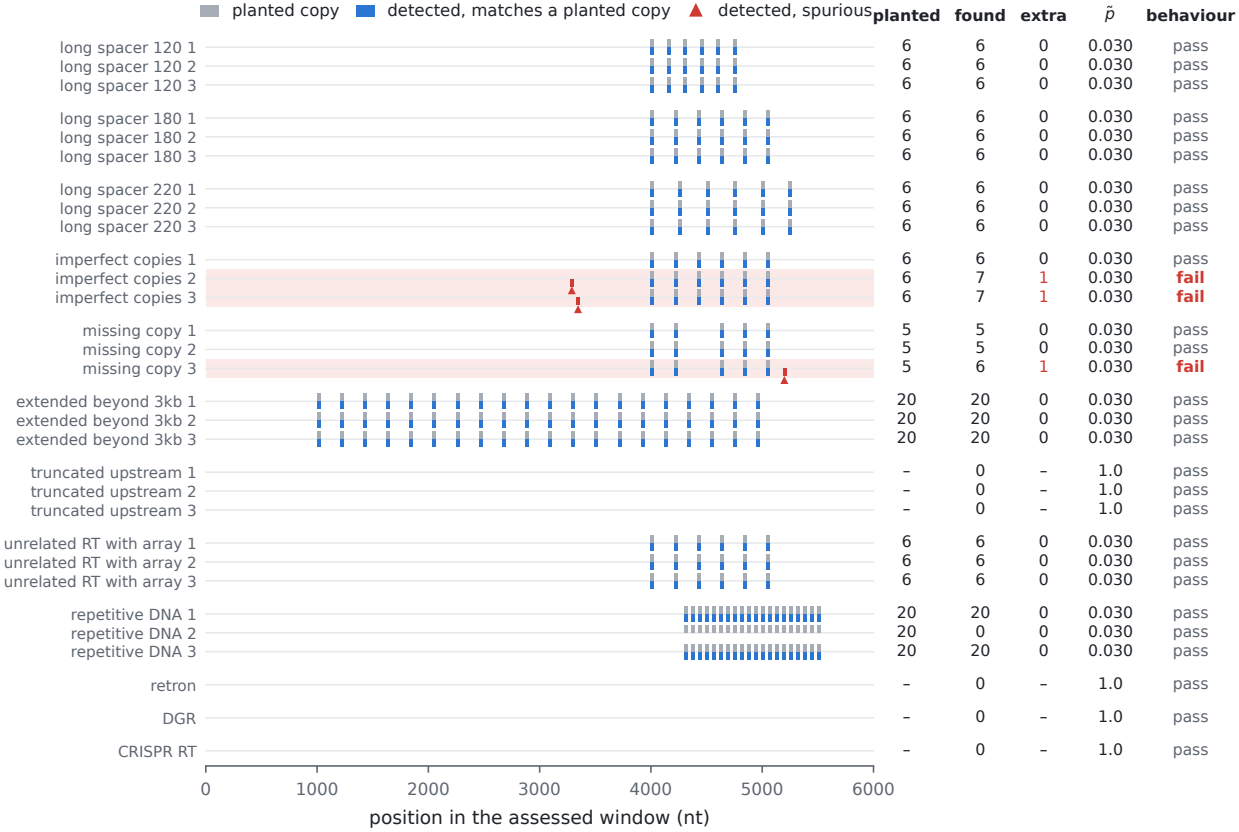
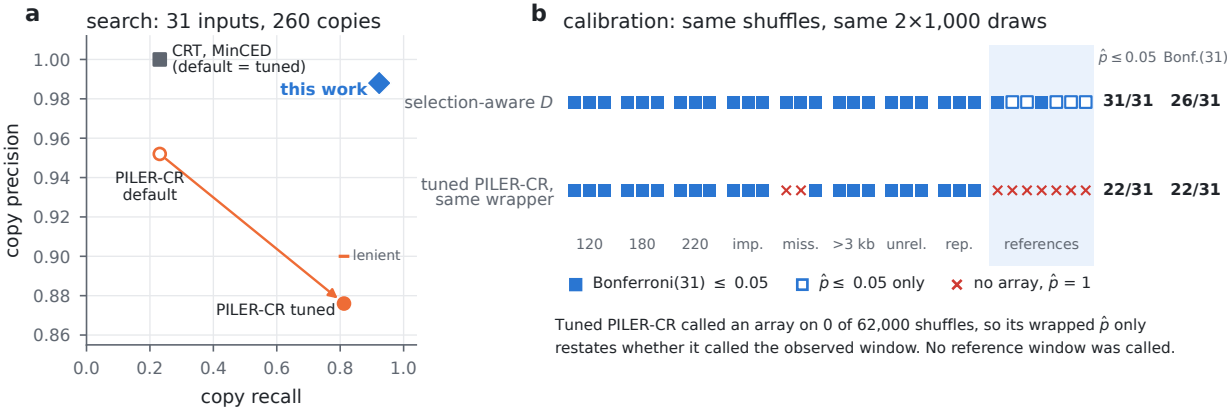
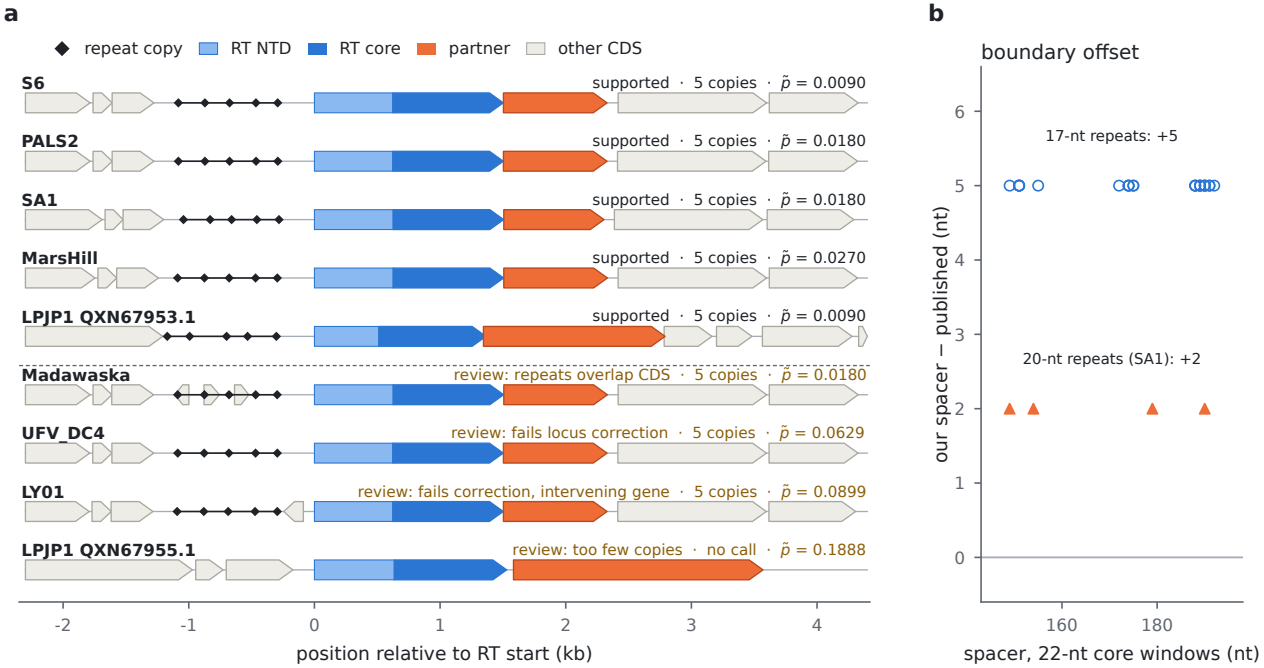
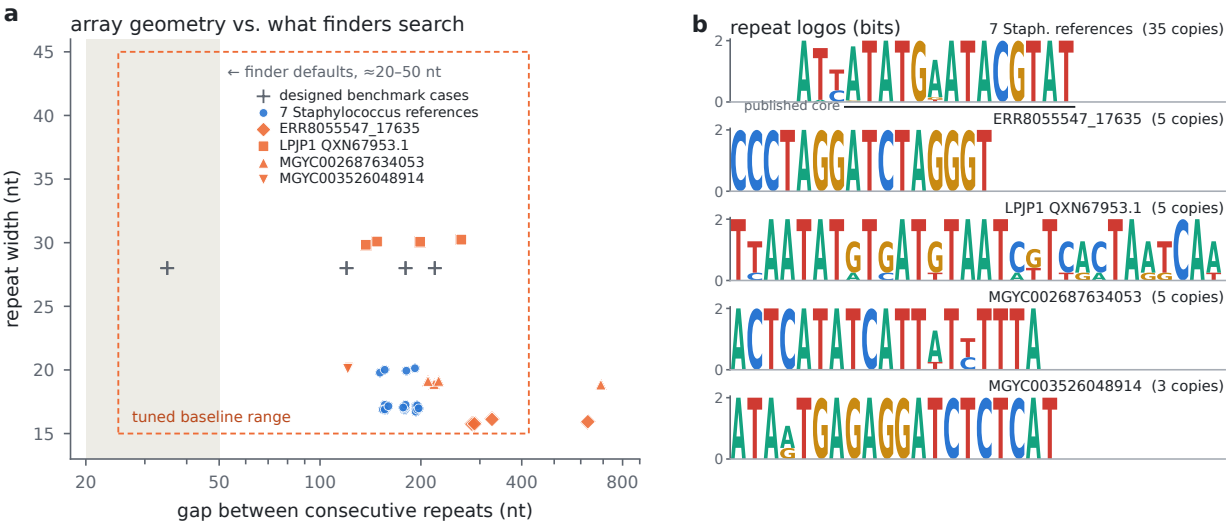

observed
locus L, 6 kb upstream of RT
null: B shuffles each
$\mathcal{M}_{\text{dinuc}}$ Euler-trail shuffle
$\mathcal{M}_{\text{block}}$ 50-nt block shuffle
two backgrounds, run independently, never pooled
the same deterministic D
1 seed enumeration
$k = 10$; no filter, no cap
2 tolerant recovery
$\leq$ 1 mismatch
3 chaining, one spacing
DP on intervals; $\leq$ 1 skip
4 boundaries, information
±80 nt; 80% conserved
5 score and argmax
$S = (R-1)\cdot I$; total order
re-executed in every replicate
search-outcome null
$S(D(L))$
0
100
200
best-chain score
real draws, LPJP1 QXN67953.1
1,000 per background
$\hat{p}_{\mathcal{M}} = \frac{1 + |\{b : S(D(X_b)) \geq S(D(L))\}|}{1 + B}$
$\hat{p} = \max(\hat{p}_{\text{dinuc}}, \hat{p}_{\text{block}})$
a tract must survive both
$\tilde{p} = \min(1, m\hat{p})$
declared family of m loci
requires $m/(B+1) \leq 0.05$
relative to the two shuffle models only: not function, not a census
non-circular evidence gate
profile evidence
partner gene context
RT-only placement
may license a family call
array: $\tilde{p}$, copies, spacers
may license an array call
protein completeness
upstream assessability
can only withhold a call

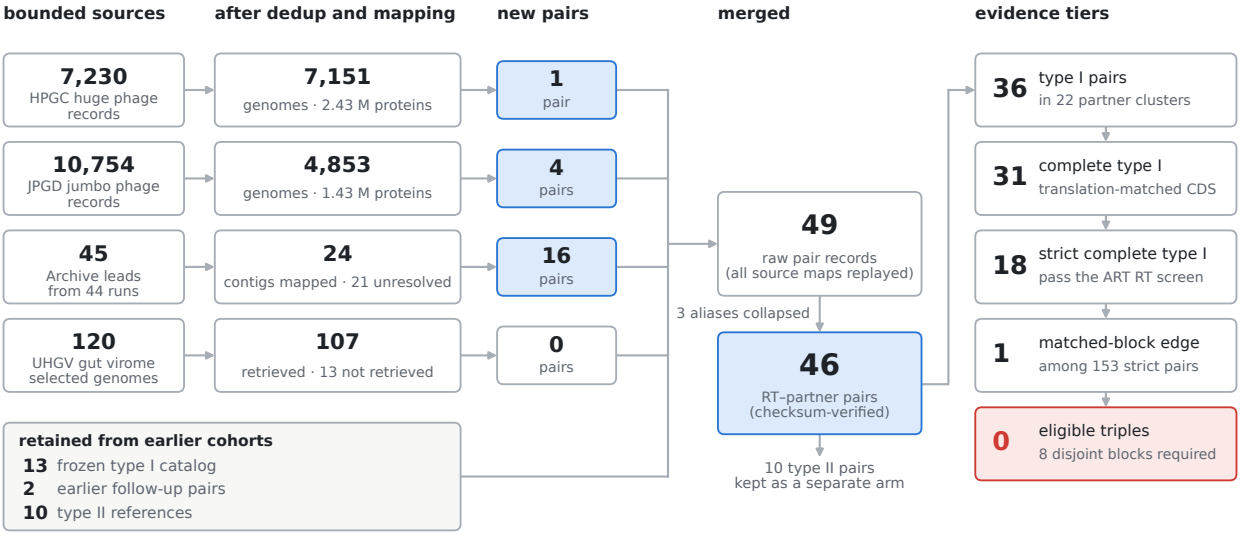

a
observed score vs. search null
long spacer 120
long spacer 180
long spacer 220
imperfect copies
missing copy
extended beyond 3kb
truncated upstream
unrelated RT with array
repetitive DNA
retron
DGR
CRISPR RT
S6
PALS2
SA1
MarsHill
Madawaska
LPJP1 QXN67953.1
LPJP1 QXN67955.1
UFV_DC4
LY01
null 5–95%
up to null max
designed
reference
100
1000
best-chain score S (log scale)
b
DGR control
observed
S = 67.4
dinucleotide
p = 0.066
local block
p = 0.003
fraction of draws ≥ s
best-chain score s
c
rejections on null draws
97.5th percentile of windows
10×
1×
rejection rate / α
other background
own background
max of the two
nominal α
d
attainable corrected p
B=200
B=1,000
B=2,000
0.05
30 cases
9 refs
17 type I
floor m/(B + 1)
declared family size m
e
7 references vs. family size
0.05
m=9: 5/7
m=31: 2/7
corrected p̃
declared family size m

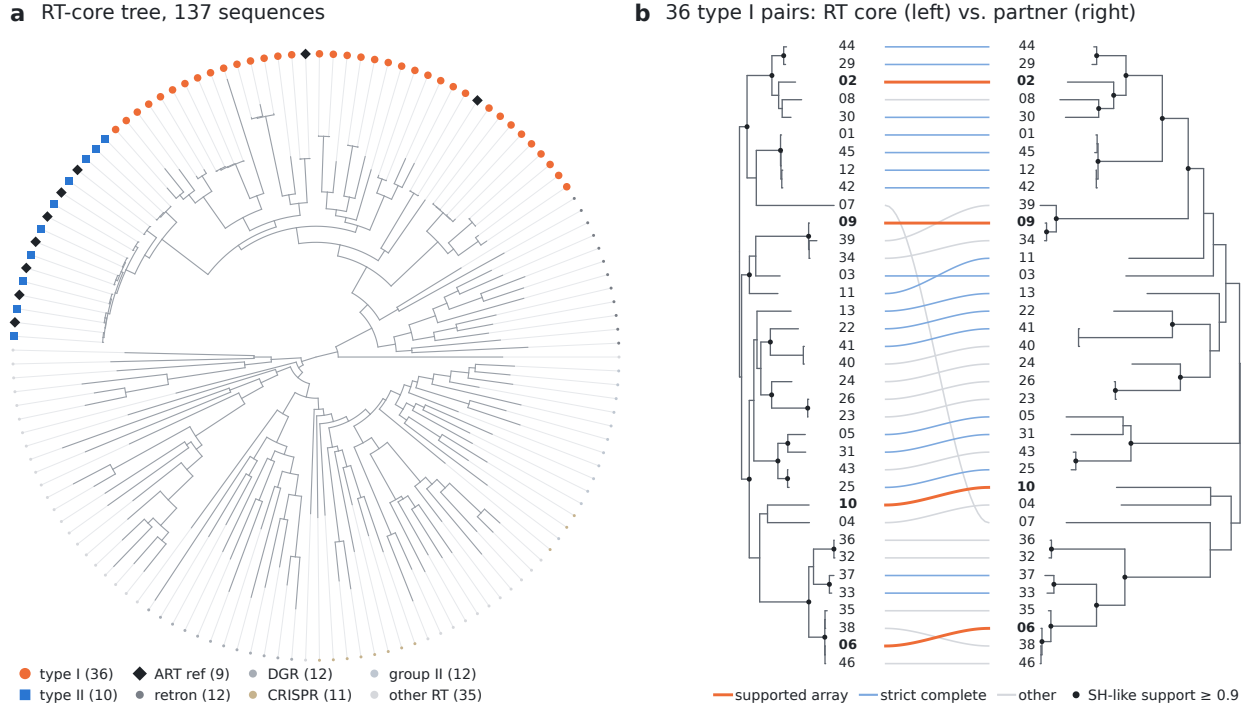

planted copy
detected, matches a planted copy
detected, spurious
planted
found
extra
$\tilde{p}$
behaviour
long spacer 120 1 6 6 0 0.030 pass
long spacer 120 2 6 6 0 0.030 pass
long spacer 120 3 6 6 0 0.030 pass
long spacer 180 1 6 6 0 0.030 pass
long spacer 180 2 6 6 0 0.030 pass
long spacer 180 3 6 6 0 0.030 pass
long spacer 220 1 6 6 0 0.030 pass
long spacer 220 2 6 6 0 0.030 pass
long spacer 220 3 6 6 0 0.030 pass
imperfect copies 1 6 6 0 0.030 pass
imperfect copies 2 6 7 1 0.030 fail
imperfect copies 3 6 7 1 0.030 fail
missing copy 1 5 5 0 0.030 pass
missing copy 2 5 5 0 0.030 pass
missing copy 3 5 6 1 0.030 fail
extended beyond 3kb 1 20 20 0 0.030 pass
extended beyond 3kb 2 20 20 0 0.030 pass
extended beyond 3kb 3 20 20 0 0.030 pass
truncated upstream 1 – 0 – 1.0 pass
truncated upstream 2 – 0 – 1.0 pass
truncated upstream 3 – 0 – 1.0 pass
unrelated RT with array 1 6 6 0 0.030 pass
unrelated RT with array 2 6 6 0 0.030 pass
unrelated RT with array 3 6 6 0 0.030 pass
repetitive DNA 1 20 20 0 0.030 pass
repetitive DNA 2 20 0 0 0.030 pass
repetitive DNA 3 20 20 0 0.030 pass
retron – 0 – 1.0 pass
DGR – 0 – 1.0 pass
CRISPR RT – 0 – 1.0 pass
0
1000
2000
3000
4000
5000
6000
position in the assessed window (nt)

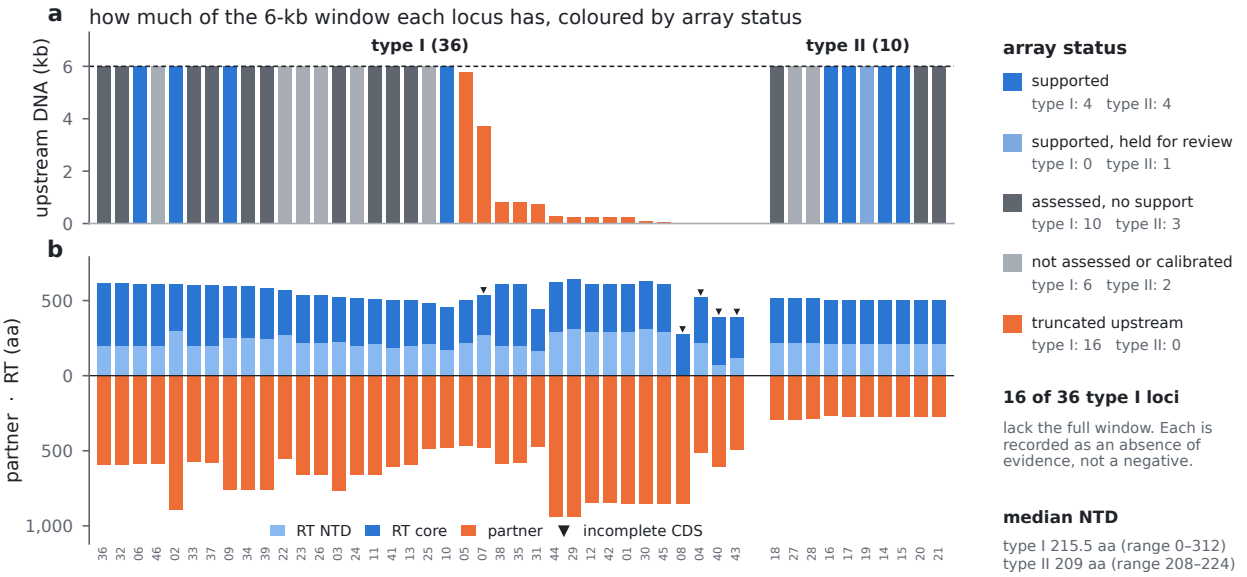

a
search: 31 inputs, 260 copies
CRT, MinCED
(default = tuned)
this work
PILER-CR
default
lenient
PILER-CR tuned
copy precision
copy recall
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.0
0.2
0.4
0.6
0.8
1.0
b
calibration: same shuffles, same 2×1,000 draws
$\hat{p} \le 0.05$ Bonf.(31)
selection-aware D
31/31
26/31
tuned PILER-CR,
same wrapper
22/31
22/31
120
180
220
imp.
miss.
>3 kb
unrel.
rep.
references
Bonferroni(31) ≤ 0.05
$\hat{p} \le 0.05$ only
no array, $\hat{p} = 1$
Tuned PILER-CR called an array on 0 of 62,000 shuffles, so its wrapped $\hat{p}$ only
restates whether it called the observed window. No reference window was called.


a
repeat copy
RT NTD
RT core
partner
other CDS
S6
supported · 5 copies · p̃ = 0.0090
PALS2
supported · 5 copies · p̃ = 0.0180
SA1
supported · 5 copies · p̃ = 0.0180
MarsHill
supported · 5 copies · p̃ = 0.0270
LPJP1 QXN67953.1
supported · 5 copies · p̃ = 0.0090
Madawaska
review: repeats overlap CDS · 5 copies · p̃ = 0.0180
UFV_DC4
review: fails locus correction · 5 copies · p̃ = 0.0629
LY01
review: fails correction, intervening gene · 5 copies · p̃ = 0.0899
LPJP1 QXN67955.1
review: too few copies · no call · p̃ = 0.1888
-2
-1
0
1
2
3
4
position relative to RT start (kb)
b
boundary offset
17-nt repeats: +5
20-nt repeats (SA1): +2
our spacer − published (nt)
0
1
2
3
4
5
6
160
180
spacer, 22-nt core windows (nt)


a
array geometry vs. what finders search
← finder defaults, ≈20–50 nt
designed benchmark cases
7 Staphylococcus references
ERR8055547_17635
LPJP1 QXN67953.1
MGYC002687634053
MGYC003526048914
tuned baseline range
repeat width (nt)
gap between consecutive repeats (nt)
b
repeat logos (bits)
7 Staph. references (35 copies)
published core
ERR8055547_17635 (5 copies)
LPJP1 QXN67953.1 (5 copies)
MGYC002687634053 (5 copies)
MGYC003526048914 (3 copies)


bounded sources
after dedup and mapping
new pairs
merged
evidence tiers
7,230
HPGC huge phage records
7,151
genomes · 2.43 M proteins
1
pair
10,754
JPGD jumbo phage records
4,853
genomes · 1.43 M proteins
4
pairs
45
Archive leads from 44 runs
24
contigs mapped · 21 unresolved
16
pairs
120
UHGV gut virome selected genomes
107
retrieved · 13 not retrieved
0
pairs
retained from earlier cohorts
13 frozen type I catalog
2 earlier follow-up pairs
10 type II references
49
raw pair records (all source maps replayed)
3 aliases collapsed
46
RT–partner pairs (checksum-verified)
10 type II pairs kept as a separate arm
36
type I pairs
in 22 partner clusters
31
complete type I
translation-matched CDS
18
strict complete type I
pass the ART RT screen
1
matched-block edge
among 153 strict pairs
0
eligible triples
8 disjoint blocks required


a
RT-core tree, 137 sequences
type I (36)
type II (10)
ART ref (9)
retron (12)
DGR (12)
CRISPR (11)
group II (12)
other RT (35)
b
36 type I pairs: RT core (left) vs. partner (right)
44
29
02
08
30
01
45
12
42
07
09
39
34
03
11
13
22
41
40
24
26
23
05
31
43
25
10
04
36
32
37
33
35
38
06
46
44
29
02
08
30
01
45
12
42
39
09
34
11
03
13
22
41
40
24
26
23
05
31
43
25
10
04
07
36
32
37
33
35
06
38
46
supported array
strict complete
other
SH-like support ≥ 0.9


a
how much of the 6-kb window each locus has, coloured by array status
type I (36)
type II (10)
upstream DNA (kb)
6
4
2
0
b
partner · RT (aa)
500
0
500
1,000
RT NTD
RT core
partner
incomplete CDS
36 32 06 46 02 33 37 09 34 39 22 23 26 03 24 11 41 13 25 10 05 07 38 35 31 44 29 12 42 01 30 45 08 04 40 43
18 27 28 16 17 19 14 15 20 21
array status
supported
type I: 4 type II: 4
supported, held for review
type I: 0 type II: 1
assessed, no support
type I: 10 type II: 3
not assessed or calibrated
type I: 6 type II: 2
truncated upstream
type I: 16 type II: 0
16 of 36 type I loci
lack the full window. Each is recorded as an absence of evidence, not a negative.
median NTD
type I 215.5 aa (range 0–312)
type II 209 aa (range 208–224)

Exploratory and partial: 56 of 1,221 requests returned; no contrast computed

